

# FACTSHEET FS03 · Momentum

**What changed on 2 October 2026.** Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS\_RERUN\_2026-10-02.md.

**US, 1999–2012 (pre-registered second period, run 2 October 2026).** The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net:  $t - 0.36$  (2013–2026:  $+0.31$ ), no 2013–2026 effect to test. Long-only alpha vs the equal-weight universe:  $+0.3\%$  a year,  $t + 0.06$  (2013–2026:  $+1.74$ ), same sign, weaker. US momentum shows nothing in 1999–2012, a period that includes the 2009 momentum crash. Registered verdicts are unchanged. Pre-registration, method and every result: PREREG\_US\_1999\_2012.md.

Buy last year's winners, sell last year's losers

Henrik Gjerming · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

|                                                             |                                                                                            |                                    |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------|
| STRATEGY                                                    | ORIGIN                                                                                     | FAMILY                             |
| Cross-sectional momentum<br>(Carhart's UMD, AQR's momentum) | Jegadeesh & Titman (1993)                                                                  | Momentum & trend                   |
| UNIVERSES                                                   | BACKTEST                                                                                   | BOOKS                              |
| US · EU · UK · DK · SCANDI · World<br>(point-in-time)       | Feb 2013 – Aug 2026, monthly, net<br>of costs                                              | Long/short and long-only (winners) |
| REBALANCE                                                   | STATUS                                                                                     |                                    |
| Monthly                                                     | Registered gate $ t  > 2.87$ over 12<br>cells; every cell in the programme<br>trial ledger |                                    |

**The claim.** Stocks that did best over the past year keep outperforming the worst over the next months. Jegadeesh & Titman (1993) found about 1% a month for US stocks in 1965–1989; Asness, Moskowitz & Pedersen (2013) found it in every major equity market and asset class. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned  $-2.1\%$  to  $+9.8\%$  a year and was positive in 5 of the six; no universe passes the 2.87 gate. The long-only book (winners) had a higher Sharpe than the equal-weight universe in 5 of the six (US, EU, UK, DK, World), with alphas of  $+0.2\%$  to  $+9.0\%$ , passing the gate in UK, World. **What goes wrong (§9):** the main drags are a market bet (average beta  $-0.38$ , worth  $-4.5\%$  a year in 2013–19), a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (appendix C):** Momentum & trend; its returns sit closest to the momentum cluster of the signal library. **360° view (appendix D):** nearest neighbours Near 52-week high, Volatility; after its closest neighbours and the market it keeps an alpha of  $-3.9\%$  to  $+7.2\%$  (largest  $|t|$  2.2), so it carries some information of its own.

Layout revised 2 October 2026 (verdict and scorecard, Sharpe anatomy, fit with the other strategies; detail moved to the appendix). No number changed.

# 1. Verdict and scorecard

**Verdict.** Of the 12 registered cells (two books × six universes), **2 pass** the  $|t| > 2.87$  gate (long-only UK, long-only World). Programme-wide (772 trials,  $|t| > 3.99$ ): nothing survives. US 1999–2012, a period the factsheet was not built on: L/S  $t$  -0.36, long-only alpha  $t$  +0.06.

| Universe     | L/S $t$ | L/S Sharpe | Long-only alpha $t$ | Long-only Sharpe vs equal-weight | Long-only max DD |
|--------------|---------|------------|---------------------|----------------------------------|------------------|
| US           | +0.31   | 0.08       | +1.74               | 0.87 vs 0.85                     | -20%             |
| EU           | +1.93   | 0.49       | +1.89               | 0.96 vs 0.75                     | -33%             |
| UK           | +1.52   | 0.41       | +3.71 ✓             | 1.13 vs 0.68                     | -25%             |
| DK           | +1.29   | 0.34       | +2.64               | 1.07 vs 0.83                     | -31%             |
| SCANDI       | -0.54   | -0.14      | +0.09               | 0.76 vs 0.89                     | -34%             |
| World        | +1.22   | 0.31       | +2.99 ✓             | 0.94 vs 0.71                     | -25%             |
| US 1999–2012 | -0.36   | -0.11      | +0.06               | 0.33 vs 0.49                     | –                |

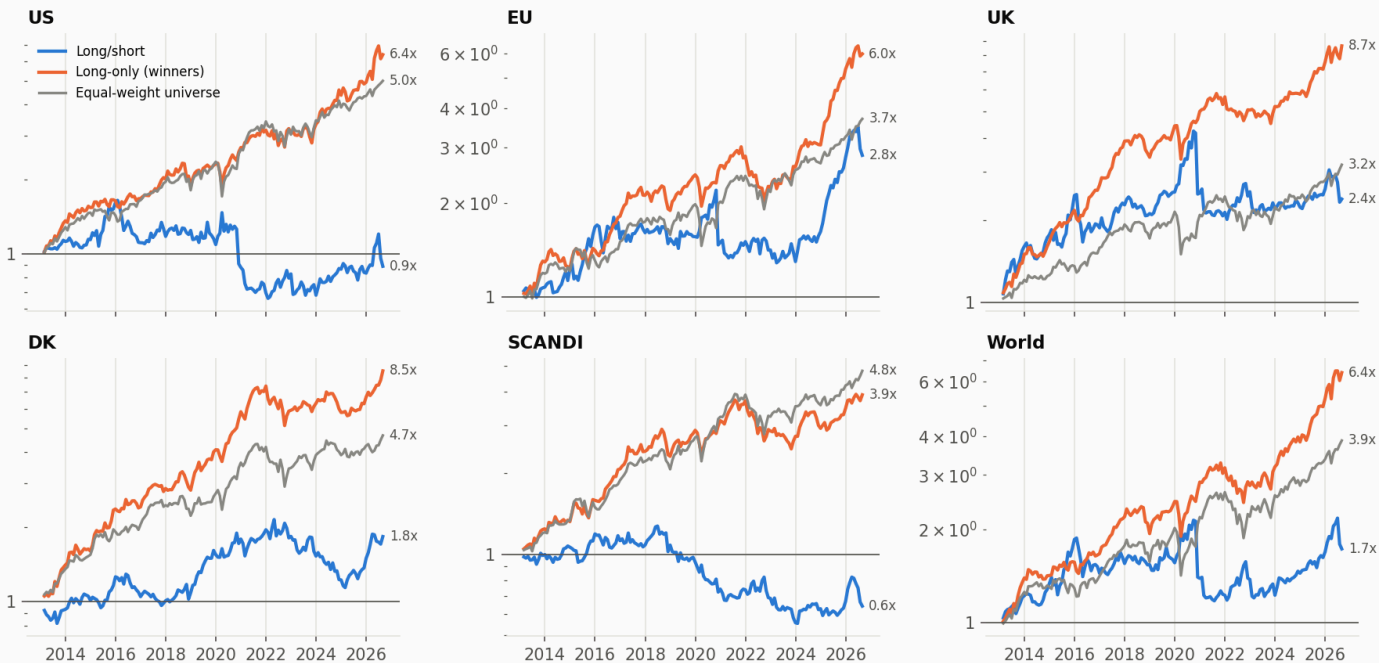
✓ passes the 2.87 gate (Bonferroni over 12 cells); ✗ significantly negative. 2013–2026 unless stated.

# 2. Headline performance

| Universe | Names / groups | L/S return / yr | L/S $t$ | L/S CAPM alpha ( $t$ ) | L/S beta | Long-only CAGR | Long-only Sharpe | Long-only max DD | EW CAGR | EW Sharpe | Size-weighted CAGR |
|----------|----------------|-----------------|---------|------------------------|----------|----------------|------------------|------------------|---------|-----------|--------------------|
| US       | 501 / 10       | +1.7%           | 0.31    | +9.8% (2.4)            | -0.62    | +14.6%         | 0.87             | -20%             | +12.6%  | 0.85      | +14.9%             |
| EU       | 264 / 10       | +9.8%           | 1.93    | +16.7% (3.6)           | -0.64    | +14.1%         | 0.96             | -33%             | +10.1%  | 0.75      | +10.2%             |
| UK       | 324 / 10       | +9.3%           | 1.52    | +15.6% (3.0)           | -0.66    | +17.2%         | 1.13             | -25%             | +8.9%   | 0.68      | +7.8%              |
| DK       | 20 / 3         | +5.9%           | 1.29    | +9.4% (2.1)            | -0.28    | +17.1%         | 1.07             | -31%             | +12.0%  | 0.83      | +10.2%             |
| SCANDI   | 70 / 5         | -2.1%           | -0.54   | +2.1% (0.6)            | -0.34    | +10.5%         | 0.76             | -34%             | +12.2%  | 0.89      | +11.0%             |
| World    | 1086 / 10      | +6.1%           | 1.22    | +12.9% (3.5)           | -0.60    | +14.6%         | 0.94             | -25%             | +10.5%  | 0.71      | +10.3%             |

Monthly, local currency (World in USD). L/S = equal-weighted winners (highest 12-1 return) minus losers (lowest 12-1 return), net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Momentum on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjerning · Project 10 factsheets

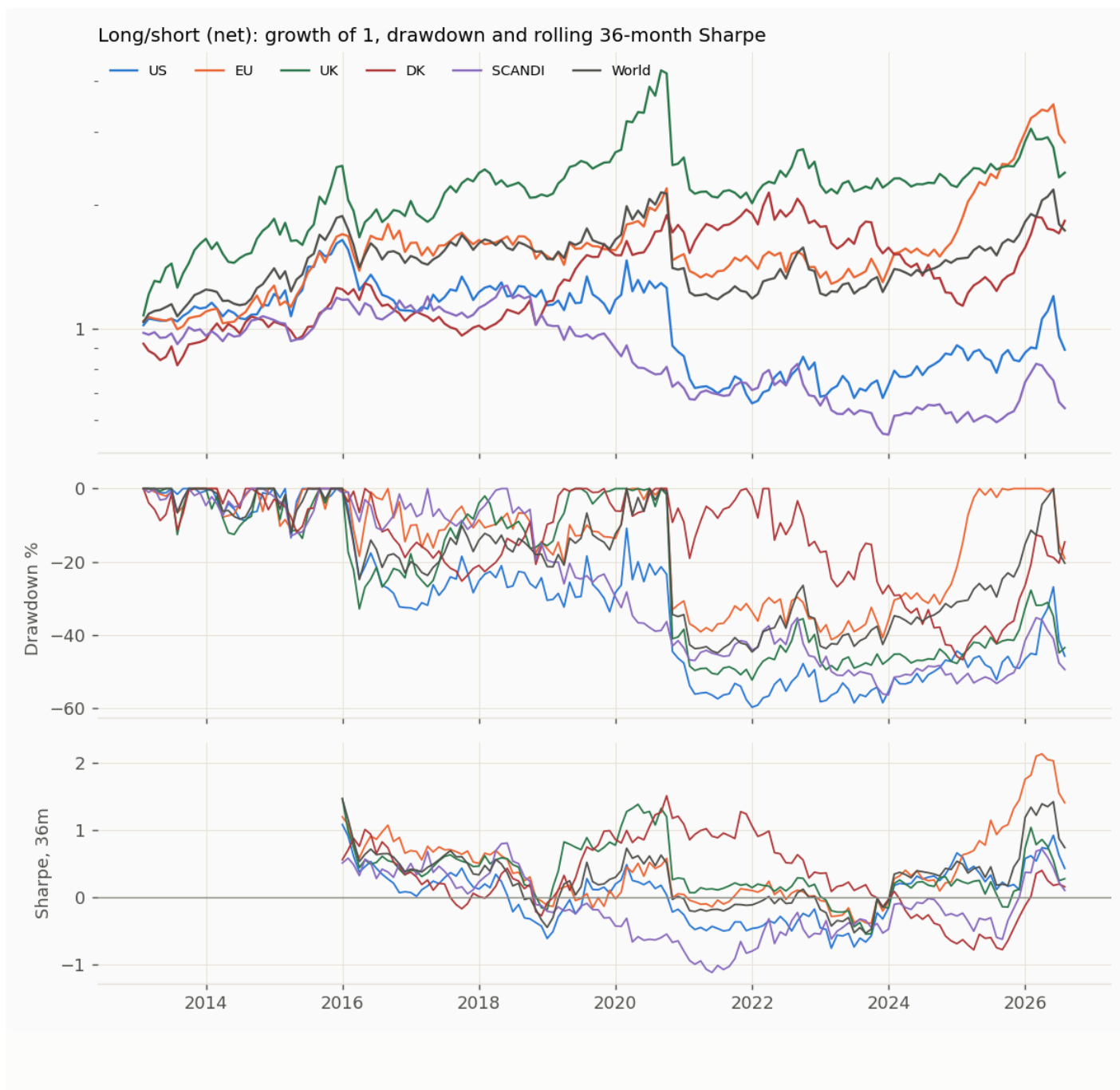
**Read with care.** A long/short book with a negative beta shows a large CAPM alpha in a rising market. The factor-model alpha in section 7, after momentum, value and the other themes, is the better guide: here the L/S CAPM alpha ranges +2.1% to +16.7% and the factor alpha -10.4% to +2.8%.

### 3. Where the Sharpe comes from

Sharpe = annual return ÷ annual volatility. The return splits into the part the market explains (beta × market return) and the rest (alpha, net of costs); the volatility into the market's share and the strategy's own. The last column is the Sharpe the book would have with its market exposure hedged out (before hedging costs).

| Universe | Book      | Return / yr | = market part | + alpha | Costs / yr (inside the return) | Volatility | of which market | Sharpe | Sharpe, market hedged |
|----------|-----------|-------------|---------------|---------|--------------------------------|------------|-----------------|--------|-----------------------|
| US       | L/S       | +1.7%       | -8.1%         | +9.8%   | -2.5%                          | 22.2%      | 9.6%            | 0.08   | 0.49                  |
| US       | Long-only | +15.2%      | +11.8%        | +3.4%   | –                              | 17.4%      | 13.9%           | 0.87   | 0.33                  |
| EU       | L/S       | +9.8%       | -6.9%         | +16.7%  | -2.2%                          | 19.9%      | 9.2%            | 0.49   | 0.94                  |
| EU       | Long-only | +14.3%      | +8.7%         | +5.6%   | –                              | 15.0%      | 11.7%           | 0.96   | 0.60                  |





## 4. How it fits next to the market and the other strategies

A long/short book improves a market portfolio when its own Sharpe beats the hurdle: its correlation with the market  $\times$  the market's Sharpe. A negative correlation makes the hurdle negative, so even a book with a small positive Sharpe diversifies; the size of the gain still depends on that Sharpe.

| Universe | Corr. with market | Avg corr. with the other strategies | Most similar strategy | Market Sharpe | Hurdle Sharpe | This L/S Sharpe | Verdict | L/S in the worst 10% of market months | Market in those months |
|----------|-------------------|-------------------------------------|-----------------------|---------------|---------------|-----------------|---------|---------------------------------------|------------------------|
| US       | -0.43             | +0.12                               | 52-week high (+0.84)  | 0.85          | -0.37         | 0.08            | adds    | +2.5%                                 | -7.3%                  |

| Universe | Corr. with market | Avg corr. with the other strategies | Most similar strategy | Market Sharpe | Hurdle Sharpe | This L/S Sharpe | Verdict | L/S in the worst 10% of market months | Market in those months |
|----------|-------------------|-------------------------------------|-----------------------|---------------|---------------|-----------------|---------|---------------------------------------|------------------------|
| EU       | -0.46             | +0.22                               | 52-week high (+0.84)  | 0.75          | -0.35         | 0.49            | adds    | +4.5%                                 | -6.5%                  |
| UK       | -0.41             | +0.30                               | 52-week high (+0.84)  | 0.68          | -0.27         | 0.41            | adds    | +3.3%                                 | -6.6%                  |
| DK       | -0.24             | +0.11                               | 52-week high (+0.66)  | 0.83          | -0.20         | 0.34            | adds    | +3.0%                                 | -7.2%                  |
| SCANDI   | -0.32             | +0.15                               | 52-week high (+0.74)  | 0.89          | -0.28         | -0.14           | adds    | +2.9%                                 | -6.9%                  |
| World    | -0.50             | +0.19                               | 52-week high (+0.88)  | 0.71          | -0.35         | 0.31            | adds    | +3.3%                                 | -7.5%                  |

*Long/short (net), monthly, 2013–2026; market = equal-weight universe; other strategies = the long/short books of FS01–FS12. The Strategy Cockpit lets you build books of several strategies.*

## 5. Strategy description

Stocks that did best over the past year keep outperforming the worst over the next months. Jegadeesh & Titman (1993) found about 1% a month for US stocks in 1965–1989; Asness, Moskowitz & Pedersen (2013) found it in every major equity market and asset class.

**Why it might work.** Behavioural: investors under-react to news and then herd (Barberis, Shleifer & Vishny 1998; Hong & Stein 1999); the disposition effect slows the price response (Grinblatt & Han 2005). Risk stories exist but explain little.

| Rule      | Original (Jegadeesh & Titman (1993))                            | This factsheet                                             |
|-----------|-----------------------------------------------------------------|------------------------------------------------------------|
| Signal    | Past 3–12 month return; the classic 12-1 skips the latest month | Return from month $t-11$ to $t-1$ (skip month $t$ )        |
| Portfolio | Decile or tercile long/short, 1–12 month holding                | Extreme quantile L/S, one-month holding, monthly rebalance |
| Weighting | Equal or value weighted                                         | Equal weighted                                             |

## 6. Method: data, signal and portfolio

### Data load

| Universe | Prices                                                                  | Membership                                                                                                                                                                                                                | Names eligible (median) | Currency                   |
|----------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|
| US       | Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted       | <b>Point-in-time:</b> the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)                                                                                                           | ~503                    | USD                        |
| EU       | Project1 yfinance cache, Adj Close (TR)                                 | <b>Point-in-time:</b> union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced | ~238                    | local (EUR, SEK, DKK, PLN) |
| UK       | Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×) | <b>Point-in-time:</b> FTSE 100 + FTSE 250, quarterly snapshots; 75% priced                                                                                                                                                | ~242                    | GBP                        |
| DK       | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMX Copenhagen 25, quarterly snapshots; 91% priced                                                                                                                                                  | ~19                     | DKK                        |
| SCANDI   | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced                                                                                                                                 | ~62                     | local (DKK, SEK, EUR)      |
| World    | US Sharadar + UK + EU above                                             | <b>Point-in-time</b> union of the three; local-currency returns summed, not converted                                                                                                                                     | ~963                    | mixed                      |

*Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.*

### Signal creation

$MOM_{i,t} = P_{i,t-1} / P_{i,t-12} - 1$  on total-return prices; the most recent month is skipped to avoid the one-month reversal. Signals use only information up to month-end  $t$  and are traded in month  $t+1$ . Eligibility: in the point-in-time universe on any of the last five trading days of month  $t$ .

### Model build

- Sort eligible stocks into equal-count groups (deciles  $\geq 100$  names, quintiles 50–99, terciles  $< 50$ ; fixed per universe).
- **Long/short** = winners (highest 12-1 return) minus losers (lowest 12-1 return); **long-only** = winners (highest 12-1 return); benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash

financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

## 7. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West  $t$ , 6 lags.

### Long/short

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with $ t  \geq 2$ (t)                                                                    |
|----------|------------------------|------------|----------|----------------|---------------------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | -4.6%      | -1.80    | 0.81           | value -0.62 (-5.4), momentum +1.51 (+11.3)                                                        |
| EU       | French Europe 5F + WML | -4.1%      | -1.42    | 0.66           | WML +1.50 (+13.8)                                                                                 |
| UK       | JKP GBR 7 themes       | +2.4%      | 0.61     | 0.68           | momentum +1.22 (+7.0), low_risk +0.94 (+3.1), quality +0.94 (+2.3)                                |
| DK       | JKP DNK 7 themes       | -3.5%      | -1.14    | 0.55           | size -0.29 (-2.7), momentum +1.14 (+11.9), quality +0.28 (+2.5), short_term_reversal +0.27 (+2.5) |
| SCANDI   | French Europe 5F + WML | -10.4%     | -2.88    | 0.35           | WML +0.85 (+7.6)                                                                                  |
| World    | JKP World 7 themes     | +2.8%      | 0.64     | 0.63           | value -0.53 (-3.3), momentum +1.26 (+9.2)                                                         |

### Long-only

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with $ t  \geq 2$ (t)                                                                           |
|----------|------------------------|------------|----------|----------------|----------------------------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | +0.9%      | 0.64     | 0.90           | mkt +0.96 (+23.5), size -0.41 (-3.8), momentum +0.79 (+9.8), low_risk -0.31 (-4.3)                       |
| EU       | French Europe 5F + WML | +3.1%      | 1.00     | 0.56           | Mkt-RF +0.75 (+13.3), WML +0.53 (+6.2)                                                                   |
| UK       | JKP GBR 7 themes       | +8.1%      | 3.78     | 0.70           | mkt +0.57 (+11.6), momentum +0.65 (+6.8), low_risk -0.59 (-4.6), quality +0.69 (+4.3)                    |
| DK       | JKP DNK 7 themes       | +5.8%      | 2.23     | 0.68           | mkt +0.68 (+15.9), size -0.18 (-2.4), momentum +0.36 (+4.0), low_risk -0.46 (-5.3), quality +0.31 (+3.2) |
| SCANDI   | French Europe 5F + WML | +2.5%      | 0.72     | 0.48           | Mkt-RF +0.65 (+9.2), WML +0.28 (+3.5)                                                                    |
| World    | JKP World 7 themes     | +6.5%      | 2.76     | 0.77           | mkt +0.69 (+11.6), momentum +0.61 (+6.9)                                                                 |

## 8. Statistical detail

- **Gate:**  $|t| > 2.87$  (2 books  $\times$  6 universes, Bonferroni 0.05/12, two-sided).

- **L/S mean:** positive in 5 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** +2.1% to +16.7%, |t| past the gate in 3 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 5 of the six; alpha +0.2% to +9.0%, passing in 2 of 6.
- **Probabilistic Sharpe** of the L/S,  $P(\text{true Sharpe} > 0)$ : US 0.61, EU 0.95, UK 0.91, DK 0.89, SCANDI 0.30, World 0.86.
- **Publication decay:** gross L/S +2.0% to +17.5% a year in 2013–19 against -2.5% to +13.8% in 2020–26.

## 9. What goes wrong, and how to fix it

**Method.** Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FSO3–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FSO3–FS12, gate  $p < 0.0020$ .

### 9.1 Diagnosis (design window 2013–2019)

| Universe | L/S gross / yr | Beta  | Beta drag / yr | CAPM alpha (gross) | Costs / yr | Long leg vs EW / yr | EW vs short leg / yr | Turnover / month | Worst months                                   |
|----------|----------------|-------|----------------|--------------------|------------|---------------------|----------------------|------------------|------------------------------------------------|
| US       | +5.5%          | -0.57 | -7.5%          | +13.0%             | -1.3%      | +0.6%               | +4.9%                | 56%              | 2015-04 -13%,<br>2016-04 -12%,<br>2016-03 -10% |
| EU       | +10.3%         | -0.58 | -6.2%          | +16.5%             | -1.3%      | +4.1%               | +6.3%                | 55%              | 2016-12 -11%,<br>2015-02 -10%,<br>2016-03 -10% |
| UK       | +17.5%         | -0.43 | -5.0%          | +22.4%             | -1.3%      | +11.5%              | +6.0%                | 52%              | 2016-02 -17%,<br>2016-04 -14%,<br>2013-08 -13% |
| DK       | +9.1%          | 0.01  | +0.2%          | +9.0%              | -0.9%      | +6.7%               | +2.5%                | 36%              | 2018-11 -13%,<br>2013-08 -10%,<br>2013-02 -8%  |
| SCANDI   | +2.0%          | -0.18 | -2.7%          | +4.8%              | -1.2%      | +1.8%               | +0.2%                | 50%              | 2018-11 -15%,<br>2015-04 -9%, 2017-05 -8%      |
| World    | +10.0%         | -0.54 | -5.8%          | +15.8%             | -1.3%      | +3.6%               | +6.4%                | 54%              | 2016-04 -12%,<br>2015-04 -11%,<br>2016-03 -8%  |

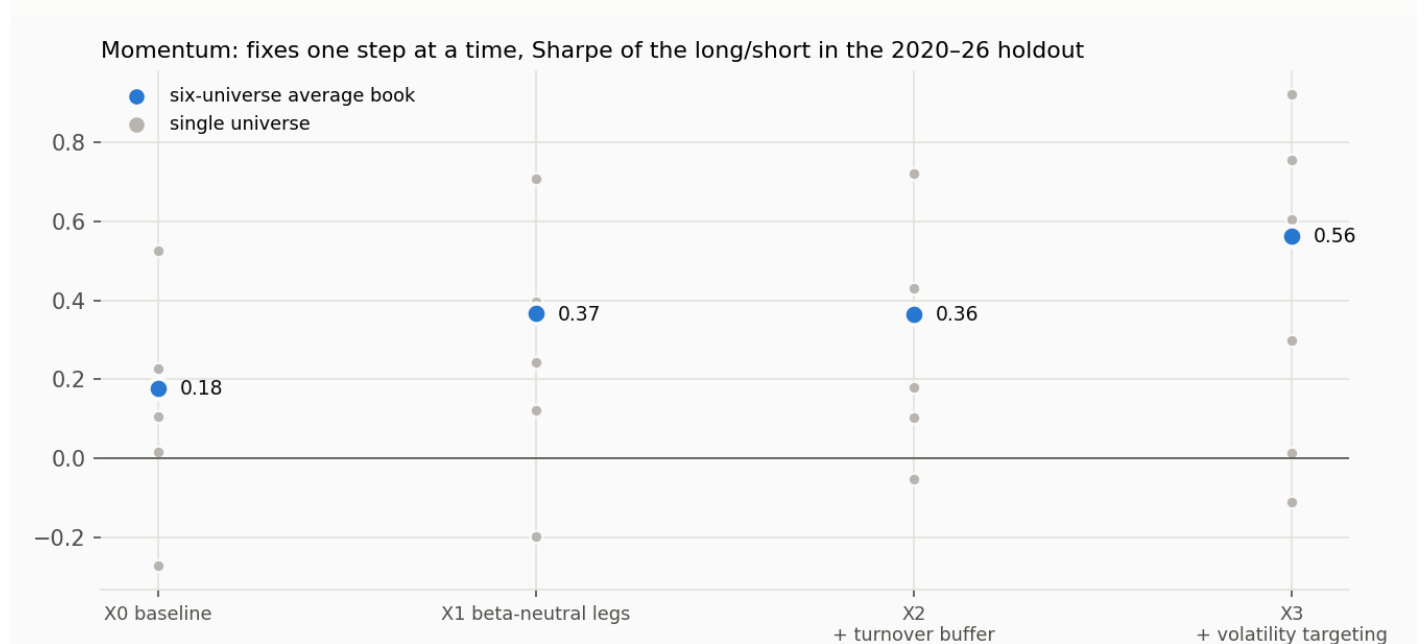
On average across the six universes the gross spread was +9.1% a year, of which the market exposure (beta -0.38) contributed -4.5%; the beta-adjusted spread (CAPM alpha) was +13.6%. Costs took 1.2% a year at 51% monthly turnover across both legs. The long leg beat the universe by +4.7% and the short leg lagged it by +4.4% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +6.4%, +6.6%, +5.9% a year.

### 9.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

| Step                      | Design Sharpe (avg) | Δ design (p)    | Holdout Sharpe (book) | Δ holdout (p)   | Universes improved | Holdout return / yr | Holdout max DD | Holdout alpha vs EW (t) | Costs / yr |
|---------------------------|---------------------|-----------------|-----------------------|-----------------|--------------------|---------------------|----------------|-------------------------|------------|
| X0 baseline               | 0.39                | –               | 0.18                  | –               | –                  | +3.2%               | -37%           | +9.7% (2.0)             | 1.2%       |
| X1 beta-neutral legs      | 0.55                | +0.23 (p 0.022) | 0.37                  | +0.19 (p 0.080) | 6 of 6             | +4.8%               | -25%           | +7.6% (1.7)             | 1.3%       |
| X2 + turnover buffer      | 0.54                | -0.02 (p 0.559) | 0.36                  | -0.00 (p 0.493) | 4 of 6             | +4.3%               | -26%           | +6.8% (1.6)             | 0.6%       |
| X3 + volatility targeting | 0.43                | -0.16 (p 0.648) | 0.56                  | +0.20 (p 0.009) | 4 of 6             | +5.8%               | -17%           | +7.9% (2.1)             | 0.6%       |

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



- **X1 beta-neutral legs:** +0.23 design, +0.19 holdout; helps in both windows but does not pass the gate.
- **X2 + turnover buffer:** -0.02 design, -0.00 holdout; hurts in both windows.
- **X3 + volatility targeting:** -0.16 design, +0.20 holdout; helps in one window and hurts in the other: not reliable.

**Where it ends:** holdout Sharpe 0.18 → 0.56, return +3.2% → +5.8% a year (six-universe book).

## 10. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Trials.** Every gated cell is in the factsheet trial ledger; the fix ladder is counted inside FS03–FS12 (25 trials).

## 11. Academic references

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## 12. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse\_xs.py FS03 → build\_xs.py FS03.  
Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.

## Appendix

### A. Performance in detail

**US (S&P 500 members, point-in-time)**, median 501 names, 10 groups

| Book                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]    | Sortino | Max DD | Hit | t(mean) | Beta  | Alpha/yr (t) | CVaR 95% (month) |
|------------------------|-----------|--------|-------|--------------------|---------|--------|-----|---------|-------|--------------|------------------|
| L/S (net)              | +1.7%     | -0.9%  | 22.2% | 0.08 [-0.39, 0.53] | -0.05   | -60%   | 53% | 0.31    | -0.62 | +9.8% (2.4)  | 15.2%            |
| Long-only (net)        | +15.2%    | +14.6% | 17.4% | 0.87 [0.49, 1.31]  | 1.46    | -20%   | 60% | 4.55    | 0.90  | +3.4% (1.7)  | 9.7%             |
| Winners (gross)        | +15.9%    | +15.4% | 17.4% | 0.92 [0.53, 1.36]  | 1.56    | -20%   | 61% | 4.77    | 0.90  | +4.1% (2.1)  | 9.6%             |
| Losers (gross)         | +11.7%    | +8.5%  | 26.7% | 0.44 [0.00, 0.93]  | 0.51    | -46%   | 58% | 1.87    | 1.52  | -8.2% (-2.9) | 15.1%            |
| EW universe            | +13.1%    | +12.6% | 15.5% | 0.85 [0.41, 1.42]  | 1.29    | -28%   | 67% | 4.20    | 1.00  | –            | 9.2%             |
| Size-weighted universe | +14.9%    | +14.9% | 14.2% | 1.05 [0.60, 1.61]  | 1.73    | -24%   | 69% | 5.29    | 0.86  | +3.7% (2.4)  | 8.3%             |

**EU (11 national blue-chip indices, point-in-time)**, median 264 names, 10 groups

| Book            | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]    | Sortino | Max DD | Hit | t(mean) | Beta  | Alpha/yr (t)  | CVaR 95% (month) |
|-----------------|-----------|--------|-------|--------------------|---------|--------|-----|---------|-------|---------------|------------------|
| L/S (net)       | +9.8%     | +8.0%  | 19.9% | 0.49 [0.01, 1.09]  | 0.56    | -41%   | 61% | 1.93    | -0.64 | +16.7% (3.6)  | 13.0%            |
| Long-only (net) | +14.3%    | +14.1% | 15.0% | 0.96 [0.37, 1.64]  | 1.54    | -33%   | 63% | 3.22    | 0.81  | +5.6% (1.9)   | 8.3%             |
| Winners (gross) | +15.0%    | +14.9% | 15.0% | 1.00 [0.42, 1.69]  | 1.65    | -32%   | 64% | 3.38    | 0.81  | +6.3% (2.1)   | 8.3%             |
| Losers (gross)  | +3.0%     | +0.3%  | 23.7% | 0.13 [-0.36, 0.58] | 0.02    | -45%   | 48% | 0.51    | 1.46  | -12.6% (-4.3) | 12.5%            |



| Book                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]     | Sortino | Max DD | Hit | t(mean) | Beta  | Alpha/yr (t) | CVaR 95% (month) |
|------------------------|-----------|--------|-------|---------------------|---------|--------|-----|---------|-------|--------------|------------------|
| L/S (net)              | -2.1%     | -3.2%  | 15.0% | -0.14 [-0.62, 0.35] | -0.28   | -56%   | 50% | -0.54   | -0.34 | +2.1% (0.6)  | 9.9%             |
| Long-only (net)        | +11.1%    | +10.5% | 14.7% | 0.76 [0.21, 1.36]   | 1.15    | -34%   | 63% | 2.83    | 0.87  | +0.2% (0.1)  | 8.4%             |
| Winners (gross)        | +11.7%    | +11.2% | 14.7% | 0.80 [0.25, 1.40]   | 1.24    | -33%   | 63% | 2.98    | 0.87  | +0.8% (0.4)  | 8.3%             |
| Losers (gross)         | +11.9%    | +10.6% | 19.2% | 0.62 [0.12, 1.18]   | 0.88    | -43%   | 58% | 2.45    | 1.21  | -3.3% (-1.4) | 11.5%            |
| EW universe            | +12.6%    | +12.2% | 14.2% | 0.89 [0.37, 1.51]   | 1.38    | -29%   | 66% | 3.63    | 1.00  | –            | 8.4%             |
| Size-weighted universe | +11.4%    | +11.0% | 13.5% | 0.84 [0.37, 1.41]   | 1.32    | -26%   | 63% | 3.59    | 0.90  | +0.0% (0.0)  | 8.1%             |

**World (US + UK + EU, point-in-time)**, median 1086 names, 10 groups

| Book                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]    | Sortino | Max DD | Hit | t(mean) | Beta  | Alpha/yr (t) | CVaR 95% (month) |
|------------------------|-----------|--------|-------|--------------------|---------|--------|-----|---------|-------|--------------|------------------|
| L/S (net)              | +6.1%     | +4.1%  | 19.4% | 0.31 [-0.16, 0.87] | 0.28    | -45%   | 58% | 1.22    | -0.60 | +12.9% (3.5) | 14.0%            |
| Long-only (net)        | +15.0%    | +14.6% | 15.9% | 0.94 [0.46, 1.51]  | 1.55    | -25%   | 63% | 4.04    | 0.86  | +5.3% (3.0)  | 9.2%             |
| Winners (gross)        | +15.7%    | +15.4% | 15.9% | 0.99 [0.51, 1.55]  | 1.65    | -25%   | 63% | 4.23    | 0.86  | +6.0% (3.4)  | 9.1%             |
| Losers (gross)         | +7.4%     | +4.2%  | 25.8% | 0.29 [-0.18, 0.77] | 0.26    | -45%   | 53% | 1.13    | 1.46  | -9.1% (-3.7) | 14.0%            |
| EW universe            | +11.3%    | +10.5% | 16.0% | 0.71 [0.25, 1.28]  | 1.02    | -30%   | 63% | 3.07    | 1.00  | –            | 9.7%             |
| Size-weighted universe | +11.0%    | +10.3% | 14.8% | 0.74 [0.28, 1.31]  | 1.10    | -28%   | 61% | 3.19    | 0.91  | +0.7% (0.8)  | 8.9%             |

*Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.*

**Calendar-year returns, L/S (net)**

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | +12.1% | +7.6%  | +60.8% | -7.0%  | -0.2%  | +22.4% |
| 2014 | +1.7%  | +13.9% | +10.2% | +8.6%  | +6.5%  | +9.5%  |
| 2015 | +41.4% | +36.5% | +39.1% | +24.6% | +12.0% | +38.6% |
| 2016 | -31.2% | -8.2%  | -23.8% | -16.8% | -5.7%  | -22.2% |
| 2017 | +6.0%  | +4.4%  | +21.3% | -3.7%  | -3.6%  | +7.6%  |
| 2018 | +5.7%  | -4.2%  | -7.0%  | +7.7%  | -0.0%  | -0.8%  |
| 2019 | -12.3% | +1.4%  | +19.8% | +42.6% | -13.4% | +1.4%  |

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2020 | -19.2% | -4.1%  | -0.9%  | +15.5% | -20.7% | -10.9% |
| 2021 | -21.1% | +0.6%  | -14.9% | +9.0%  | -0.5%  | -11.8% |
| 2022 | +19.3% | -5.9%  | +18.9% | -6.0%  | -7.1%  | +12.4% |
| 2023 | -18.0% | -8.2%  | -13.5% | -16.8% | -18.8% | -11.1% |
| 2024 | +27.4% | +24.6% | +1.5%  | -19.9% | +12.7% | +17.7% |
| 2025 | -3.6%  | +72.6% | +16.4% | +22.3% | +6.6%  | +17.1% |
| 2026 | +6.6%  | +1.5%  | -8.0%  | +22.3% | -4.1%  | +2.3%  |

Calendar-year returns, long-only (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | +35.9% | +29.4% | +50.6% | +40.4% | +24.3% | +39.7% |
| 2014 | +13.0% | -4.7%  | +11.1% | +13.7% | +6.3%  | +1.8%  |
| 2015 | +6.5%  | +14.4% | +30.1% | +47.9% | +20.0% | +10.4% |
| 2016 | +2.2%  | +28.4% | +23.5% | +2.0%  | +30.7% | +6.5%  |
| 2017 | +19.8% | +16.6% | +44.7% | +18.7% | +16.2% | +31.4% |
| 2018 | -6.4%  | -10.5% | -12.7% | -1.6%  | -4.2%  | -11.7% |
| 2019 | +22.9% | +30.0% | +30.2% | +45.6% | +25.0% | +26.9% |
| 2020 | +16.6% | +7.1%  | +13.0% | +42.3% | +8.5%  | +15.5% |
| 2021 | +17.5% | +7.1%  | +12.9% | +27.0% | +18.6% | +11.1% |
| 2022 | -3.0%  | -20.5% | -10.5% | -20.2% | -22.8% | -12.3% |
| 2023 | +8.9%  | +16.3% | +3.1%  | +6.7%  | -4.4%  | +14.9% |
| 2024 | +31.0% | +17.5% | +10.8% | -8.2%  | +7.0%  | +20.2% |
| 2025 | +12.9% | +74.6% | +31.0% | +15.5% | +18.8% | +38.0% |
| 2026 | +29.2% | +11.8% | +14.6% | +27.4% | +12.1% | +21.2% |

B. Trading record and current book

The rebalance log per universe is in `results/xs_FS03_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

| Universe | Rebalances | Names long / short | Turnover long leg / month | Turnover short leg / month | L/S months positive | Worst L/S month | When    |
|----------|------------|--------------------|---------------------------|----------------------------|---------------------|-----------------|---------|
| US       | 163        | 51 / 50            | 30%                       | 26%                        | 53%                 | -27.6%          | 2020-11 |
| EU       | 163        | 27 / 26            | 29%                       | 24%                        | 61%                 | -33.1%          | 2020-11 |
| UK       | 163        | 33 / 32            | 28%                       | 26%                        | 61%                 | -40.1%          | 2020-11 |

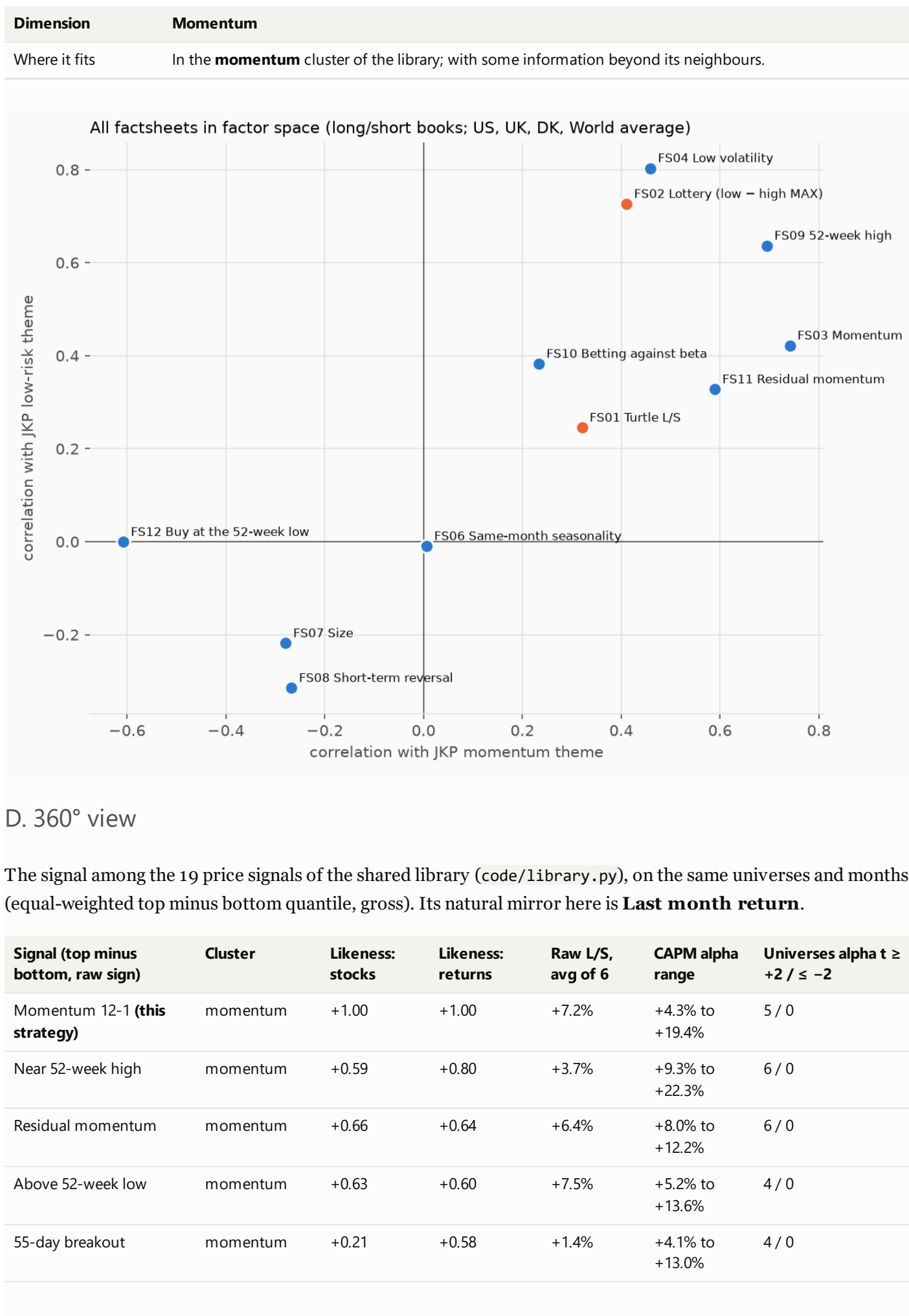
| Universe | Rebalances | Names long / short | Turnover long leg / month | Turnover short leg / month | L/S months positive | Worst L/S month | When    |
|----------|------------|--------------------|---------------------------|----------------------------|---------------------|-----------------|---------|
| DK       | 163        | 7 / 6              | 17%                       | 19%                        | 55%                 | -14.1%          | 2022-05 |
| SCANDI   | 163        | 14 / 14            | 25%                       | 24%                        | 50%                 | -14.8%          | 2018-11 |
| World    | 163        | 109 / 108          | 29%                       | 25%                        | 58%                 | -34.0%          | 2020-11 |

**Current book** (signal at 31 August 2026; first 8 names per leg)

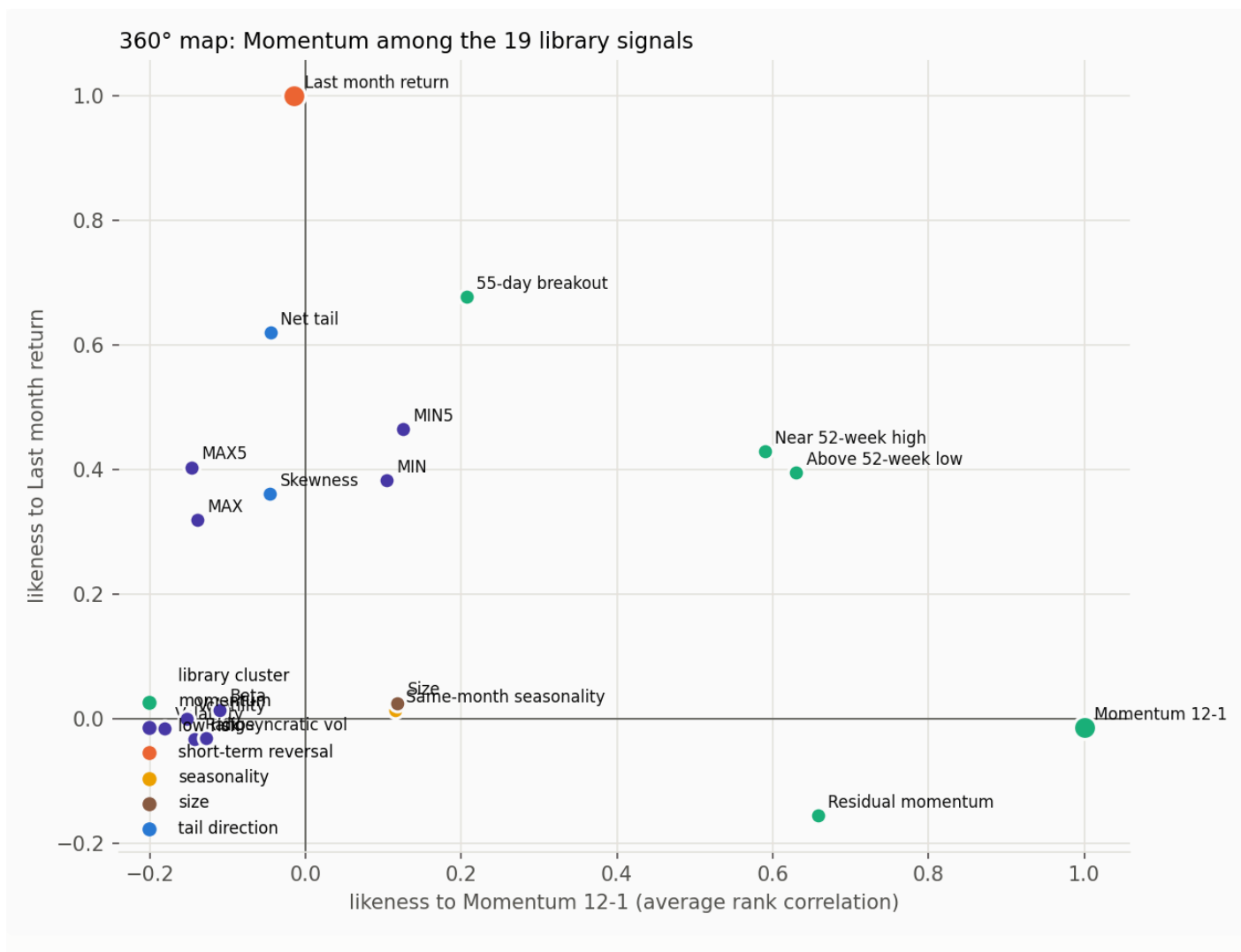
| Universe | Long: winners                                                                   | Short: losers                                                                  |
|----------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| US       | SNDK, MU, WDC, LITE, STX, CIEN, INTC, DELL                                      | CSGP, TTD, FISV, BSX, BLDR, INTU, COIN, PDD                                    |
| EU       | KGH.WA, ASML.AS, MT.AS, MTS.MC, REP.MC, ASM.AS, TYRES.HE, NOKIA.HE              | ONTEX.BR, AGFB.BR, QTCOM.HE, UMG.AS, GTC.WA, BAR.BR, ADYEN.AS, STLAM.MI        |
| UK       | SAGA.L, CMCX.L, CWR.L, KLR.L, WOSG.L, SSIT.L, CCC.L, APN.L                      | VTY.L, AML.L, TEP.L, MEGP.L, BCG.L, DATA.L, ONT.L, ENT.L                       |
| DK       | DANSKE.CO, ISS.CO, JYSK.CO, NKT.CO, VWS.CO, MAERSK-B.CO, NDA-DK.CO, MAERSK-A.CO | ZEAL.CO, COLO-B.CO, ORSTED.CO, BAVA.CO, AMBU-B.CO, GN.CO, ROCK-B.CO, NOVO-B.CO |
| SCANDI   | NOKIA.HE, TYRES.HE, NESTE.HE, DANSKE.CO, JYSK.CO, SWED-A.ST, SAND.ST, ISS.CO    | QTCOM.HE, ORSTED.CO, COLO-B.CO, BAVA.CO, PNDORA.CO, AMBU-B.CO, GN.CO, ELISA.HE |

## C. Classification: where does it fit?

| Dimension                         | Momentum                                                                                                                                                                                                                       |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Series family                     | <b>Momentum &amp; trend</b>                                                                                                                                                                                                    |
| Academic style                    | Cross-sectional momentum (Carhart's UMD, AQR's momentum)                                                                                                                                                                       |
| Signal                            | Price only (daily total returns)                                                                                                                                                                                               |
| Cross-section vs time series      | <b>Cross-sectional:</b> stocks ranked against each other each month                                                                                                                                                            |
| Horizon and turnover              | One month; 51% of the two legs replaced monthly                                                                                                                                                                                |
| Market exposure                   | L/S beta -0.66 to -0.28                                                                                                                                                                                                        |
| Payoff shape                      | Treynor–Mazuy $\gamma$ -3.73 to 0.43 (t -2.2 to 0.5); worst 10% of market months +2.5% to +4.5% a month, best 10% -6.1% to -2.2%                                                                                               |
| Nearest factor theme              | momentum (4 of 6 universes), WML (2 of 6 universes)                                                                                                                                                                            |
| Nearest library signals (returns) | Near 52-week high (6 universes), Residual momentum (4 universes), 55-day breakout (2 universes)                                                                                                                                |
| Economic rationale                | Behavioural: investors under-react to news and then herd (Barberis, Shleifer & Vishny 1998; Hong & Stein 1999); the disposition effect slows the price response (Grinblatt & Han 2005). Risk stories exist but explain little. |
| Publication                       | Jegadeesh & Titman 1993 (data to 1989); Carhart 1997 factor; known crash risk after market rebounds (Daniel & Moskowitz 2016)                                                                                                  |



| Signal (top minus bottom, raw sign) | Cluster             | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha $t \geq +2 / \leq -2$ |
|-------------------------------------|---------------------|------------------|-------------------|-------------------|------------------|---------------------------------------|
| Volatility                          | low risk            | -0.18            | -0.54             | +1.1%             | -13.7% to -7.9%  | 0 / 5                                 |
| MIN5                                | low risk            | +0.13            | +0.52             | -0.9%             | +2.3% to +13.7%  | 4 / 0                                 |
| Volatility (252 days)               | low risk            | -0.15            | -0.50             | +2.4%             | -12.9% to -4.8%  | 0 / 3                                 |
| Range (MAX–MIN)                     | low risk            | -0.14            | -0.50             | -0.7%             | -15.4% to -4.8%  | 0 / 4                                 |
| Beta                                | low risk            | -0.11            | -0.47             | +4.4%             | -12.6% to -4.2%  | 0 / 4                                 |
| MIN (worst day)                     | low risk            | +0.10            | +0.47             | -0.7%             | +3.3% to +11.1%  | 4 / 0                                 |
| MAX (best day)                      | low risk            | -0.14            | -0.42             | +0.1%             | -13.3% to -2.0%  | 0 / 2                                 |
| Idiosyncratic vol                   | low risk            | -0.13            | -0.42             | -3.6%             | -15.5% to -4.8%  | 0 / 4                                 |
| MAX5                                | low risk            | -0.15            | -0.41             | +0.6%             | -11.3% to -3.5%  | 0 / 2                                 |
| Size                                | size                | +0.12            | +0.33             | +0.1%             | -3.8% to +9.9%   | 1 / 0                                 |
| Last month return (mirror)          | short-term reversal | -0.01            | +0.33             | +1.8%             | -3.2% to +11.0%  | 2 / 0                                 |
| Same-month seasonality              | seasonality         | +0.12            | +0.14             | -1.9%             | -9.4% to +6.6%   | 0 / 1                                 |
| Skewness                            | tail direction      | -0.05            | -0.10             | -0.4%             | -6.7% to +3.9%   | 0 / 2                                 |
| Net tail (MAX+MIN)                  | tail direction      | -0.04            | +0.06             | -0.9%             | -5.2% to +5.2%   | 0 / 0                                 |



**Spanning.** Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t     | R <sup>2</sup> | Neighbours with  t  ≥ 2                                                                                                                         |
|----------|-----------------------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| US       | -2.7%                       | -1.48 | 0.89           | Near 52-week high +0.71 (+7.5), Size +0.41 (+5.4), 55-day breakout -0.35 (-4.0), Above 52-week low +0.35 (+5.5), Residual momentum +0.26 (+3.3) |
| EU       | -1.6%                       | -0.53 | 0.80           | Near 52-week high +1.07 (+9.9), Residual momentum +0.33 (+3.7), MIN5 -0.31 (-2.6)                                                               |
| UK       | +7.2%                       | 2.19  | 0.76           | Near 52-week high +1.25 (+9.9), MIN5 -0.38 (-2.2), 55-day breakout -0.28 (-2.3)                                                                 |
| DK       | -0.2%                       | -0.07 | 0.74           | Near 52-week high +0.31 (+5.8), Above 52-week low +0.41 (+6.9), Residual momentum +0.41 (+8.8), Size +0.19 (+3.4)                               |
| SCANDI   | -3.9%                       | -1.51 | 0.69           | Near 52-week high +0.67 (+9.3), Residual momentum +0.32 (+5.3), Above 52-week low +0.22 (+2.3)                                                  |
| World    | +2.4%                       | 1.26  | 0.90           | Near 52-week high +1.17 (+15.4), 55-day breakout -0.26 (-3.0), Residual momentum +0.40 (+5.5), MIN5 -0.25 (-2.9)                                |

3×3 sort, next-month return a year  
(US, EU, UK, SCANDI, World averaged)

|                    |                       |        |                        |
|--------------------|-----------------------|--------|------------------------|
| low Momentum 12-1  | 10.9%                 | 9.1%   | 8.1%                   |
| middle             | 11.1%                 | 11.6%  | 12.0%                  |
| high Momentum 12-1 | 13.0%                 | 13.8%  | 13.0%                  |
|                    | low Last month return | middle | high Last month return |

Holding Last month return fixed (down a column), moving from low to high Momentum 12-1 changes the return by +2.2, +4.7 and +4.9 points a year; holding Momentum 12-1 fixed (along a row), moving from low to high Last month return changes it by -2.8, +0.9 and +0.0. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -4% vs +27% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +7.2%, +6.4%, +6.6%, +5.9% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)

